

Unsupervised Anomaly Detection in Network Traffic: A Comparative Study of Isolation Forest and Convolutional Autoencoders

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1. Introduction and Related Work

Problem Statement. Network intrusion detection systems (NIDS) face a fundamental challenge: detecting novel attack patterns not seen during training. Traditional supervised approaches require extensive labeled data for each attack type, becoming impractical as attack vectors evolve rapidly. We address this by framing intrusion detection as an unsupervised anomaly detection problem, where models learn representations of normal traffic and flag deviations as potential threats.

Motivation. The asymmetry between benign and malicious traffic in real-world networks motivates unsupervised learning. Collecting comprehensive attack samples is expensive and often infeasible, while normal traffic is abundant. Furthermore, zero-day attacks by definition cannot be anticipated in supervised training sets. An effective unsupervised approach would detect both known and novel attack patterns by learning the manifold of normal behavior.

2. Proposed Method

Our framework implements two complementary unsupervised approaches with unified preprocessing to enable fair comparison.

2.1. Data Preprocessing Pipeline

We standardize feature extraction across both methods:

Feature Engineering. From raw UNSW-NB15 CSV files [1], we remove non-numeric categorical features (source/destination IPs, protocol names, service identifiers) that introduce sparsity. The remaining 42 numeric features capture flow statistics: packet counts, byte distributions, inter-arrival times, and protocol flags.

Normalization Strategy. We fit a StandardScaler exclusively on normal traffic samples (Label=0) from the training set. This critical design choice prevents attack patterns from biasing the normalization statistics. All subsequent

data training attacks and evaluation sets are transformed using these learned parameters.

Train-Eval Split. Following data extraction, we isolate the top-5 most frequent attack categories for training (80% split), reserving remaining attacks as unseen classes for evaluation. The evaluation set combines these held-out attacks with an equal proportion of normal traffic to assess false positive rates.

2.2. Isolation Forest Approach

We implemented a method that partitions the feature space through randomized decision trees. Each tree recursively splits data along random feature thresholds until samples are isolated. Anomalies, being sparse and distinct, require fewer splits resulting in shorter path lengths. We configure:

- `n_estimators=300` for ensemble stability
- `contamination=0.01` assuming 1% training outliers
- `max_features=1.0` to utilize full feature dimensionality

The anomaly score $s(x)$ is derived from average path length across trees, normalized by expected length under random partitioning. We set the detection threshold at the 99th percentile of training scores, classifying samples exceeding this as anomalies.

2.3. Convolutional Autoencoder

We designed this deep learning approach treating the 42-dimensional feature vector as a 1D sequence. Architecture details:

Encoder. Three convolutional blocks (Conv1D $\rightarrow$ Batch-Norm $\rightarrow$ ReLU $\rightarrow$ MaxPool) progressively downsample the input: $42 \rightarrow 21 \rightarrow 10 \rightarrow 5$ spatial dimensions with channels expanding (1, 16, 32, 64). This compressed representation feeds a linear bottleneck projecting to a 64-dimensional latent space.

Decoder. Mirroring the encoder, three transposed convolutions upsample the latent code back to the original 42-

feature space. We employ linear interpolation to align output dimensions precisely.

Training Protocol. The model minimizes per-sample MSE reconstruction loss using Adam optimizer ($\text{lr}=10^{-3}$) with gradient clipping (max norm=5.0) for 20 epochs. Only normal traffic trains the model. Anomaly threshold is set at the 95th percentile of training reconstruction errors.

3. Dataset and Benchmark

UNSW-NB15 Dataset.[1] Created by the Cyber Range Lab at UNSW Canberra, this dataset contains 2.5M network flow records with 49 features and 10 attack categories: Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms.

Preprocessing Statistics. After feature filtering, our training set contains 175,341 normal traffic samples. The top-5 attack classes comprise 82% of all attacks (150,238 samples). Evaluation set balances 37,560 unseen attacks with 37,560 normal samples totaling 75,120 test instances.

Evaluation Metrics. We report accuracy, precision, recall, and F1-score, then we analyze false positive rate (FPR) on normal traffic and true positive rate (TPR) on novel attacks separately, as these have different operational costs in production systems.

4. Experimental Results

Table 1. Performance comparison on UNSW-NB15 evaluation set

Method	Acc.	Prec.	Rec.	F1
Conv Autoencoder	0.9400	0.9440	0.9400	0.9399
Isolation Forest	0.7953	0.8026	0.7953	0.7941

4.1. Quantitative Analysis

Table 1 reveals a striking reversal of expected performance: the convolutional autoencoder substantially outperforms Isolation Forest, achieving 94.0% accuracy compared to 79.5%. This 14.5% improvement, coupled with superior precision (0.944 vs 0.803) and F1-score (0.940 vs 0.794), demonstrates the deep learning approach’s effectiveness for this feature space.

False Positive Analysis. The autoencoder achieves a remarkably low false positive rate of 1.25% (169/13,501 normal samples misclassified), while Isolation Forest produces 28.2% FPR (3,808/13,501). In operational security contexts, the autoencoder’s 22-fold reduction in false alarms represents transformative improvement, as each false positive consumes analyst time and erodes trust in the detection system.

Attack Detection. Both models demonstrate strong attack detection capabilities, with the autoencoder achieving

89.3% recall versus Isolation Forest’s 87.3%. However, the autoencoder’s superior precision (98.6% vs 75.6%) means that when it flags an attack, the alert is correct 98.6% of the time, substantially reducing alert fatigue.

4.2. Parameter Influence and Observations

Threshold Calibration. The autoencoder’s anomaly threshold (0.080155 MSE) was set at the 95th percentile of training reconstruction errors, while Isolation Forest used the 95th percentile of anomaly scores (0.544672).

Latent Dimensionality. Our 64-dimensional latent space provides sufficient capacity to encode the 42-feature input while maintaining generalization. The reconstruction threshold of 0.080 suggests the model learned a compact, discriminative representation for normal traffic patterns.

Surprising Result. The autoencoder’s dominant performance contradicts our initial hypothesis that tree-based methods would excel on engineered features. Deep analysis reveals that while features are aggregated statistics, their correlational structure benefits from the convolutional architecture’s ability to capture local dependencies. The 1D convolutions effectively learn which feature combinations characterize normal versus anomalous behavior.

Isolation Forest Limitations. Despite 300 estimators, Isolation Forest’s 28% FPR suggests the random partitioning strategy struggles with the high-dimensional feature space where normal traffic occupies complex, non-convex regions. The contamination parameter (0.01) may be mismatched to the actual data distribution.

5. Conclusions

This project let us know that convolutional autoencoders significantly outperform Isolation Forest for unsupervised anomaly detection in network intrusion datasets, achieving 94% accuracy with a 1.25% false positive rate. The 14.5% performance advantage demonstrates that deep learning architectures can effectively capture correlational patterns in engineered network features, contrary to conventional tree-based methods for tabular data.

Future Work. Investigation of ensemble methods combining both approaches may leverage complementary strengths. Additionally, attention mechanisms could identify which feature correlations drive anomaly decisions, enhancing interpretability for security analysts.

Team Contributions. Samuele Pagnotta designed the preprocessing pipeline, feature engineering and dataset split strategy. Rideewitage Lachitha Sangeeth Perera implemented and optimized the Isolation Forest approach. Mattia Prisco developed the convolutional autoencoder architecture and training protocol. All members contributed equally to evaluation of the models, report and presentation preparation.

References

- [1] <https://www.kaggle.com/datasets/mrwellsdavid/unsw-nb15/data>